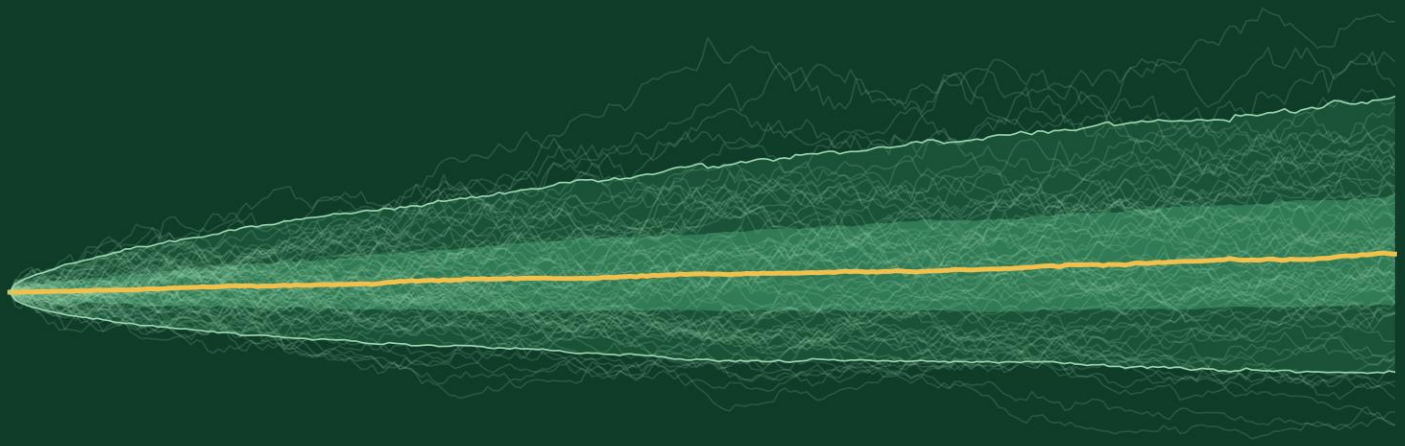
2,000 paths · 252 sessions · seed 42



FOREIGN TRADE UNIVERSITY
FACULTY OF BANKING AND FINANCE

PERSONAL PROJECT REPORT

Finfolio 2.0

Portfolio Attribution & Risk Analytics

STUDENT	Nguyễn Ngọc Anh
STUDENT ID	2413380008
INDIVIDUAL ROLE	UI/UX Designer / Information Architecture Lead
GROUP	G09 - K63
COURSE	Information Technology Applications in Finance and Banking (NHA408E)
INSTRUCTOR	Phan Trần Trung Dũng

HANOI · OCTOBER 2026

Abstract .

This report presents my individual contribution to Finfolio 2.0 as UI/UX Designer and Information Architecture Lead. The project required complex portfolio outputs to be communicated to retail investors without reducing them to an unexplained score or placing every control on a single, continuous dashboard. My work focused on the seven-page information architecture, the Forest/Paper design system, Vietnamese-compatible typography, actual-value portfolio charts, attribution presentation, responsive layout, accessibility, and the progressive disclosure of methods and limitations.

The contribution links visual hierarchy to analytical meaning: capital, return, contribution, risk, benchmark evidence, and optional research tools appear in a deliberate sequence. The report draws on application screenshots and repository artifacts to demonstrate the design decisions and their implementation, and it distinguishes interface responsibility from quantitative ownership. The Monte Carlo method and its calculation belong to the Lead Developer; my role was to maintain a coherent visual and interaction system around verified outputs. The resulting product supports desktop and mobile use while preserving units, uncertainty, and interpretive context.

KEYWORDS: *user experience; information architecture; financial visualization; responsive design; accessibility; design system.*

Contents .

I INTRODUCTION AND INDIVIDUAL RESPONSIBILITY 3

II DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0 4

2.1 Evaluation of Finfolio 1.0 and Resulting Design Changes 4

2.2 Seven-Page Information Architecture 6

2.3 Actual-Value PnL and Analytical Visualization 8

2.4 Responsive and Accessible Interaction 10

III IMPLEMENTATION AND DEVELOPMENT PROCESS 10

3.1 From Early Wireframes to a Production System 10

3.2 Visual Integration and Review 11

IV EVIDENCE, VERIFICATION AND PROJECT OUTCOMES 11

V LIMITATIONS AND LESSONS LEARNED 12

VI CONCLUSION 12

— INDIVIDUAL DECLARATION 13

List of Figures

Figure 1. Finfolio 2.0 system architecture and responsibility hand-offs 4

Figure 2. Finfolio 1.0 landing page and primary navigation structure 4

Figure 3. Color palettes of Finfolio 1.0 and Finfolio 2.0 5

Figure 4. Finfolio 2.0 landing page: initial draft (English version) 5

Figure 5. Finfolio 2.0 landing page: final design and primary navigation structure 6

Figure 6. Desktop analysis hierarchy showing result, metrics, attribution controls, and evidence 7

Figure 7. Information board providing wider explanations and results 8

Figure 8. Portfolio value and PnL results 9

Figure 9. Responsive analysis page at a mobile viewport 10

I INTRODUCTION AND INDIVIDUAL RESPONSIBILITY

Portfolio analysis interfaces are prone to information overload. A user may encounter allocation charts, performance series, risk measures, attribution tables, benchmark statistics, and optimization controls on the same screen, with no indication of which result to read first. Finfolio 2.0 addressed this problem by treating information order as part of the analytical method: the interface guides the user from an observable result to its drivers, its interpretation, possible action, and limitations.

As UI/UX Designer and Information Architecture Lead, I defined this sequence and the visual system through which it is expressed. My responsibilities covered page structure, navigation, typography, spacing, color semantics, responsive behavior, chart interaction, table containment, keyboard behavior, and the presentation of explanatory text.

The role did not include ownership of financial formulas or simulation algorithms. Hoàng Khánh Linh specified the financial meaning, Trần Minh Ngọc specified data labels and provenance, Lê Bảo An implemented the numerical engine, and Nguyễn Quỳnh Anh defined acceptance behavior. I translated their verified outputs into a coherent experience and reviewed whether the interface preserved units, assumptions, and uncertainty.

> **WHY THE BOUNDARY MATTERS**
>
> *A technically functioning application can still be methodologically invalid if its data conditions, assumptions, or explanatory boundaries are ignored.*

The collaboration followed clearly defined boundaries. Hoàng Khánh Linh approved the financial definitions, assumptions, and permissible claims. Trần Minh Ngọc defined source provenance, currency, benchmark, and compatibility requirements. Nguyễn Quỳnh Anh converted these requirements into acceptance and regression cases. Lê Bảo An implemented and integrated the specifications without silently changing their meaning. My own responsibility was to establish the information hierarchy and the visual system. Figure 1 summarizes these roles and hand-offs.

METHODOLOGY **Hoàng Khánh Linh** Approved financial definitions, assumptions and permissible claims	DATA **Trần Minh Ngọc** Defined source provenance, currency, benchmark and compatibility requirements
QUALITY ASSURANCE **Nguyễn Quỳnh Anh** Converted requirements into acceptance and regression cases	INTERFACE **Nguyễn Ngọc Anh** Established the information hierarchy and visual system

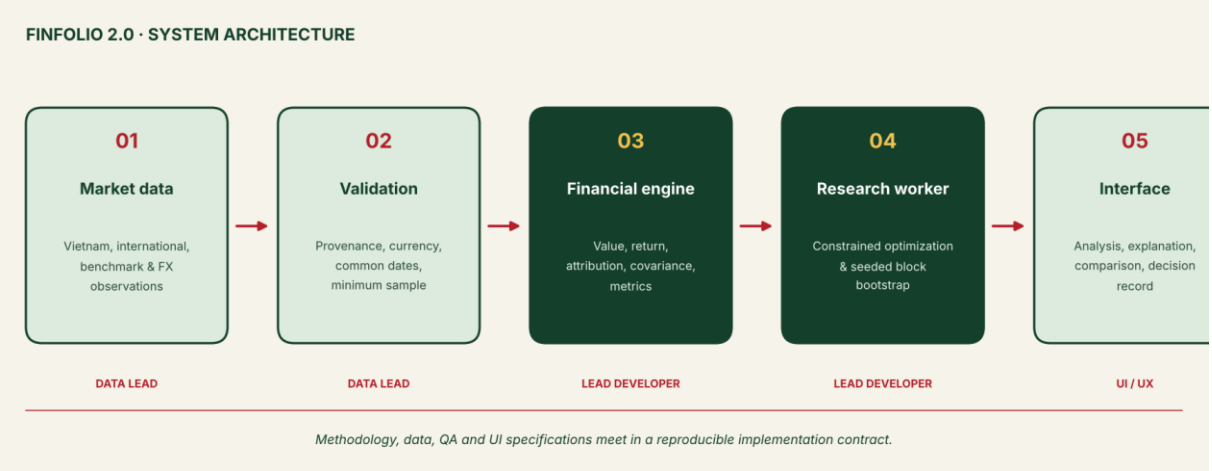


Figure 1. Finfolio 2.0 system architecture and responsibility hand-offs
Source: Author's diagram based on the Finfolio 2.0 repository.

II DISTINCTIVE CONTRIBUTIONS TO FINFOLIO 2.0

2.1 Evaluation of Finfolio 1.0 and Resulting Design Changes

As UI/UX Designer, I took on the task of evaluating the previous version of Finfolio so that the team could deliver a stronger product. This evaluation identified the key changes to be carried into Finfolio 2.0.

Overall interface

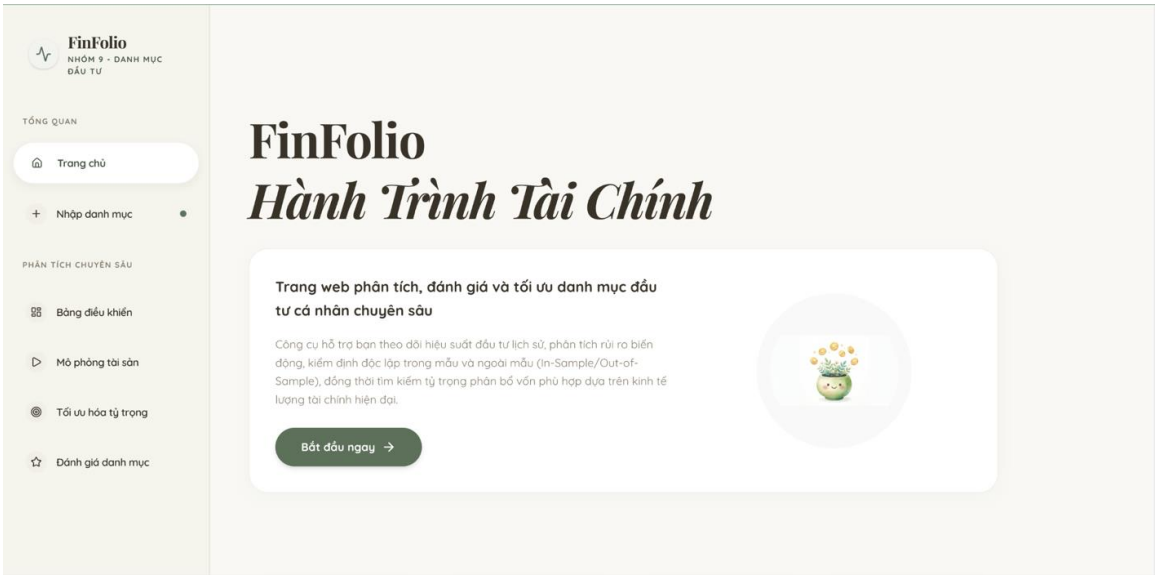


Figure 2. Finfolio 1.0 landing page and primary navigation structure
Source: Finfolio application, project screenshot.

Finfolio 1.0 lacked a color scheme that could help it stand out. Its overall appearance was a bland, empty grey with a few hints of green, and the website had no signature visual identity. The **"flower pot" mascot** was also unrelated to the purpose of the website and misleading: it suggested a cozy, domestic product rather than a financial analysis tool. Finfolio 2.0 addressed both issues.

I removed the plant mascot and introduced a new color scheme that remains simple and recognizably linked to the original site, but uses a **bolder green palette** and a more polished finish. The palette gives the product a signature identity and a more contemporary look.

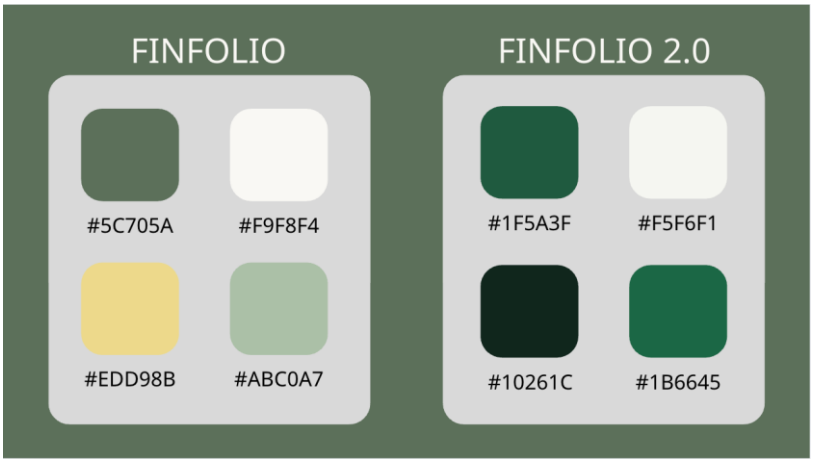


Figure 3. Color palettes of Finfolio 1.0 and Finfolio 2.0
Source: Created by the author.

I developed the Forest/Paper visual direction as a restrained system for financial information. Dark forest green identifies the product and primary actions, warm paper backgrounds reduce glare, and neutral borders organize content without turning every analytical result into a decorative card. Positive, negative, and warning states combine color with text or sign, so meaning never depends on hue alone. The interface also avoids presenting red and green as unqualified recommendations; they indicate observed contribution or validation status.

To reinforce the product's professional character, I also selected its typefaces, Fraunces and Be Vietnam. Combining the new palette and typefaces with the removal of the mascot and the criticisms identified above, I designed an entirely new interface for the website. It keeps the core color of the original (green) while presenting a more polished overall look, improving the product without erasing its history.

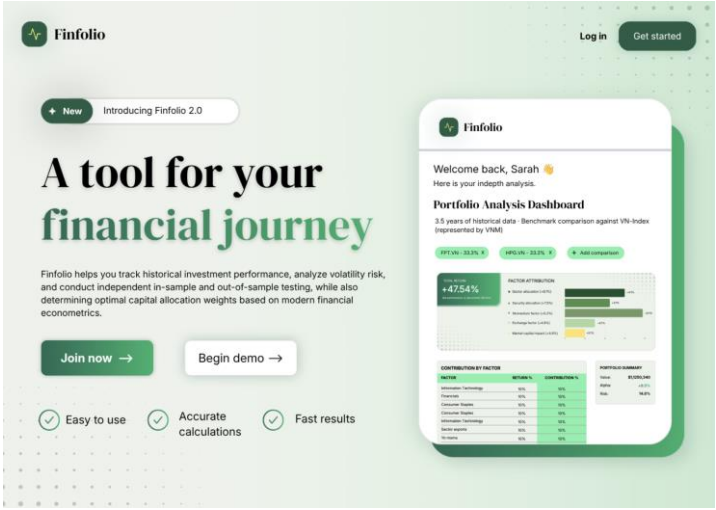


Figure 4. Finfolio 2.0 landing page: initial draft (English version)
Source: Created by the author.

After reviewing the draft with my teammates and agreeing on the main interface, we arrived at the final Finfolio 2.0 design shown in Figure 5.

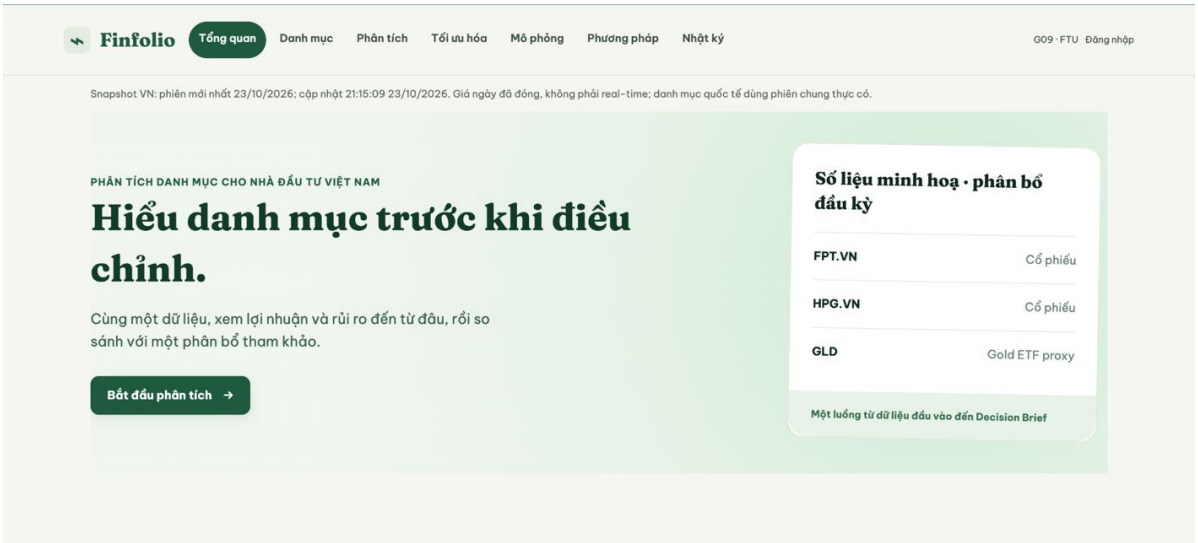


Figure 5. Finfolio 2.0 landing page: final design and primary navigation structure
Source: Created by the author.

2.2 Seven-Page Information Architecture

Information hierarchy: addressing clutter and missing structure

The previous version of Finfolio presented clustered results with too little explanation of the numbers the website calculated. The result was overwhelming and difficult to navigate, especially for users who were new to the website or to finance. Nothing in the layout indicated which results were most significant or should be read first.

For Finfolio 2.0, the final application separates overview, portfolio entry, analysis, optimization, simulation, methods, and account or decision records. This structure replaced an earlier long-page approach in which users had to scroll through input, analysis, and research features in one continuous sequence. Separating the routes reduced visual density and made the application state easier to understand. It also allowed each advanced feature to carry its own assumptions and limitations without overwhelming the core attribution report.

Within the analysis route, the hierarchy begins with the effective period, data status, and base currency. The main result then states the change in actual portfolio value. Summary metrics appear before the contribution views, followed by portfolio and benchmark charts, risk evidence, and expandable methodology. This order follows the pattern Result, Reason, Meaning, Action, and Limitation, so users can identify the result before interpreting the evidence behind it.

The navigation system uses consistent route labels and an active-state treatment on desktop. On narrow screens, it avoids forcing the full desktop navigation into a horizontal row that would overflow the viewport. The route structure also lets users return to their portfolio without losing the conceptual distinction between editing inputs and reading a report.

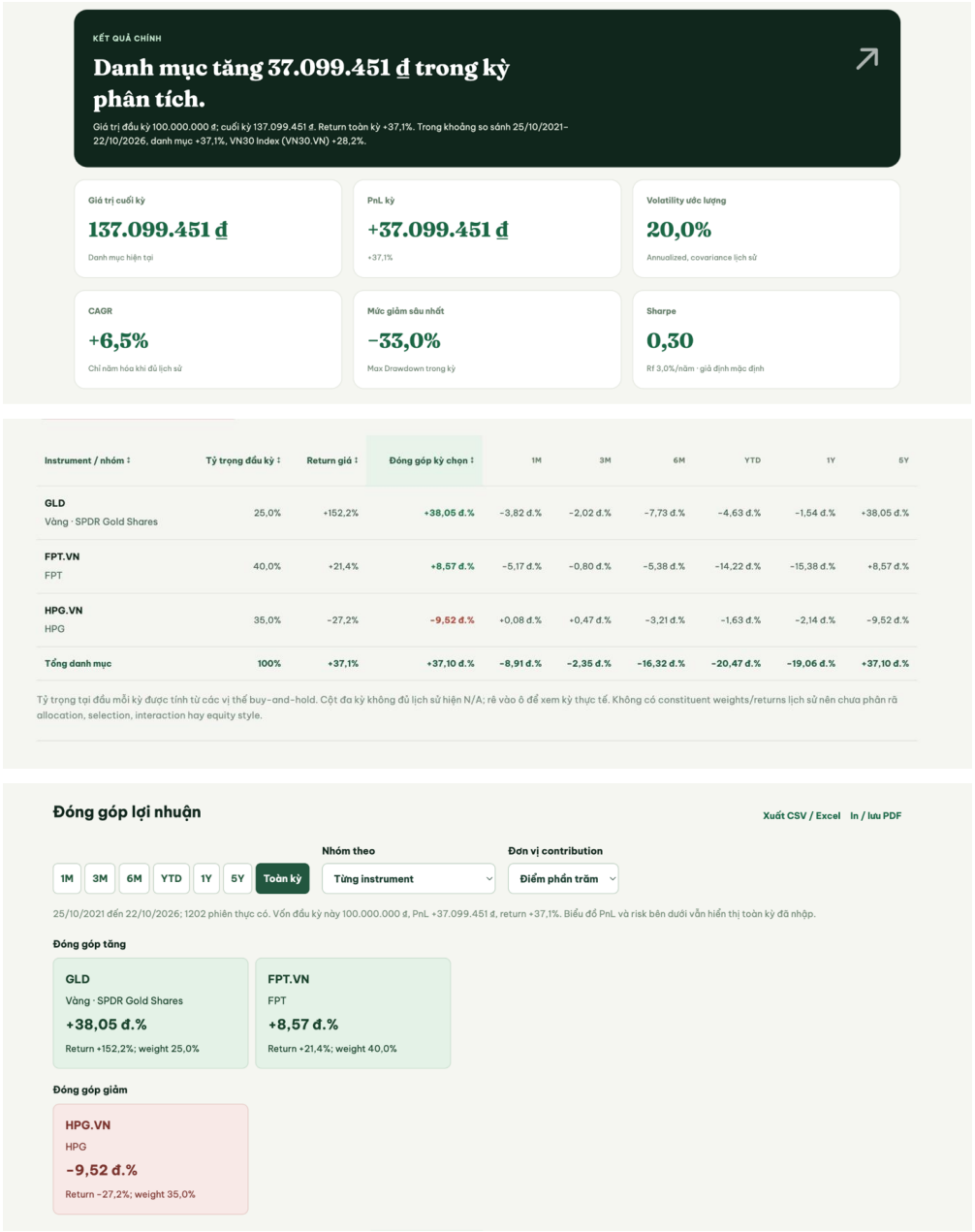


Figure 6. Desktop analysis hierarchy showing result, metrics, attribution controls, and evidence
Source: Finfolio 2.0 application, project screenshot.

Added feature: explanatory information board

Finfolio is intended for all kinds of investors, from professionals to beginners. In my assessment, however, the previous version was not welcoming to new investors: beyond the clutter of information and numbers, it offered very few explanations. I initially proposed two distinct modes (beginner and professional), but this would have overcomplicated the website and made it less economical to maintain. Because investors differ in both needs and skills, I instead proposed **a new feature: an information section that provides explanations on demand.** Professional users can continue to use the website in its most streamlined form, while newer users can open explanations when they need them, without the clutter that overwhelmed users of the previous Finfolio.

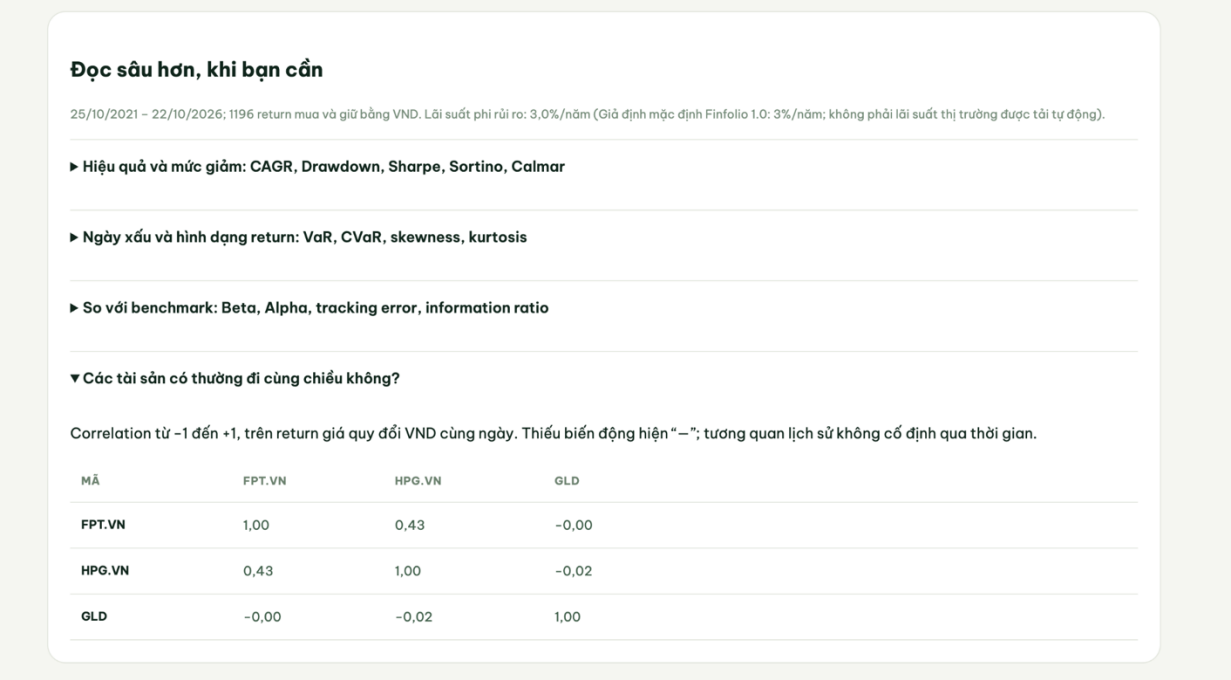


Figure 7. Information board providing wider explanations and results
Source: Finfolio 2.0 application, project screenshot.

2.3 Actual-Value PnL and Analytical Visualization

The primary performance chart is built on actual portfolio value rather than a base-100 index. This preserves the user's initial capital and gives the vertical axis an economic meaning. A VND 100 million portfolio, for example, is shown as a monetary path, and the tooltip reports the date, portfolio value, benchmark-equivalent value, and absolute PnL. The benchmark and reference-allocation lines start from the same capital, so their paths can be compared without implying that the user actually held those alternatives.

Chart interaction includes a crosshair that follows both horizontal and vertical position, an edge-aware tooltip, visible time labels, and keyboard movement between observations. Line style and labels distinguish the portfolio, benchmark, and reference allocation in addition to color. This supports users who cannot reliably distinguish the chosen hues and preserves meaning when the chart is printed in grayscale.

Attribution is visualized through ranked positive and negative contributions, tables, and grouped views. Long sector and industry labels wrap within bounded containers, and wide tables scroll horizontally on small screens. Users may group results by instrument, asset class, market or commodity type, currency, sector, or industry; the interface presents one grouping at a time to avoid visual conflict.

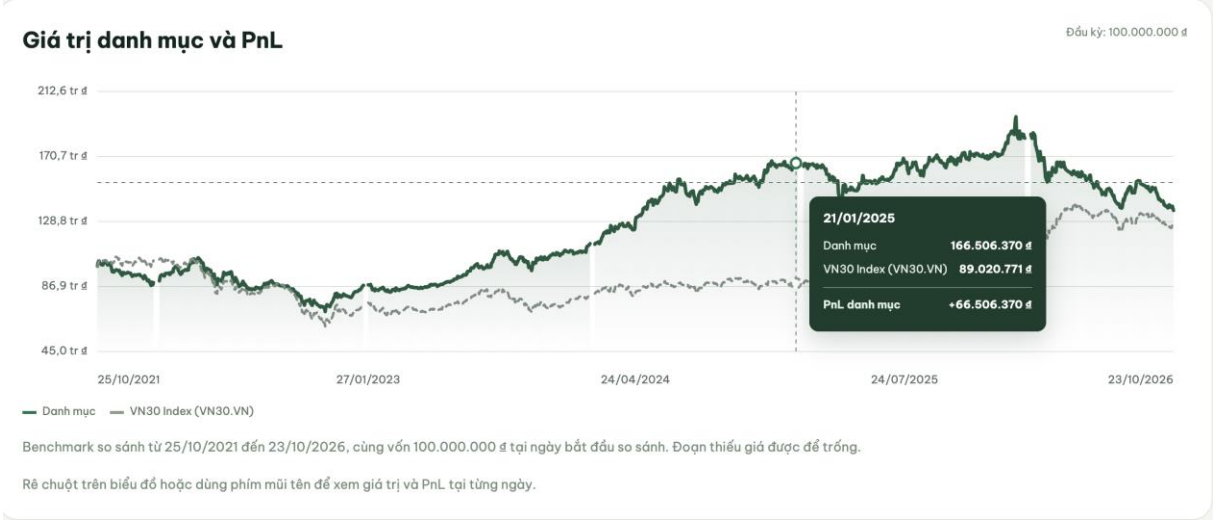


Figure 8. Portfolio value and PnL results
Source: Finfolio 2.0 application, project screenshot.

2.4 Responsive and Accessible Interaction

The previous version of Finfolio could not be displayed properly on mobile devices. For Finfolio 2.0, I worked with our developer, Lê Bảo An, to create a mobile interface and make the product more convenient to use.

Responsive behavior was reviewed at representative widths of 375, 768, and 1440 pixels. On mobile, content collapses into a single column, chart and table containers remain within the viewport, and touch targets retain usable spacing. The typography scale reduces gradually while the order of information remains stable. Figure 9 shows how the analysis heading, period, actions, source disclosure, and warnings are preserved on a narrow screen.

Accessibility work included visible keyboard focus, arrow-key navigation in the autocomplete, Escape-key behavior, descriptive labels, and non-color indicators. The interface does not hide assumptions behind icon-only controls, and advanced outputs keep explanatory text nearby. These measures do not replace formal testing with assistive-technology users, but they establish a stronger baseline than pointer-only interaction.



Figure 9. Responsive analysis page at a mobile viewport
Source: Finfolio 2.0 application, project screenshot.

III IMPLEMENTATION AND DEVELOPMENT PROCESS

3.1 From Early Wireframes to a Production System

During Weeks 1 and 2, I identified the target user's main information need and drafted the hierarchy around capital summary, performance, contribution, and risk. The early wireframes prioritized outputs that help users understand their portfolio over decorative market widgets. During Week 5, interface samples were reviewed against the core task, and controls implying brokerage execution or automated advice were excluded.

The final implementation moved from visual reference to reusable code. Design tokens provide shared color, spacing, radius, border, and elevation values, while typography rules define consistent roles rather than styling each page independently. Page-level CSS handles the specific composition of the portfolio, analysis,

optimization, simulation, method, and account views. This organization allows a change to the common system to propagate across the product while route-specific layouts are retained.

3.2 Visual Integration and Review

During Week 7, the chart utilities and analytical tables were integrated with production data. Particular attention was given to hover behavior, axis spacing, the placement of captions and methodology notes, and overflow at the lower edge of charts. Visual review found cases in which axis labels overlapped explanatory text; space was then reserved beneath the plotting area so that notes remain readable at desktop and mobile widths.

The same review addressed font compatibility and deployment rendering. A successful local build was not sufficient, because missing font files or an incorrect Vercel output configuration could change the interface in production. The final audit therefore compared deployed pages with local screenshots and verified that Vietnamese diacritics, navigation, tables, and charts retained their intended structure.

IV EVIDENCE, VERIFICATION AND PROJECT OUTCOMES

The design contribution can be reproduced directly in the application. The screenshots in this report are project evidence rather than external illustrations. A reviewer can navigate the routes at desktop and mobile widths, use the chart tooltip and keyboard controls, and compare the visible styles against the token and typography definitions.

DESIGN OUTPUT	REPOSITORY EVIDENCE	VERIFICATION PROCEDURE
Information architecture	`demo/index.html`, `demo/navigation.mjs`	Navigate the seven routes and confirm active-state and state continuity.
Design system	`demo/tokens.css`, `demo/redesign.css`	Inspect shared color, spacing, border, and layout tokens.
PnL visualization	`demo/chart-utils.mjs`	Inspect the actual-value axis, tooltip, crosshair, and line distinctions.
Responsive behavior	`demo/product.css`, `demo/redesign.css`	Compare 375, 768, and 1440 pixel viewports.

The final result is an interface in which technical sophistication does not dictate reading order. Users encounter the current portfolio result before optional optimization or simulation, while units, data timing, and limitations remain visible. This organization supports the project's problem-first direction and gives each analytical extension a clear place without collapsing the application into a single dashboard.

V LIMITATIONS AND LESSONS LEARNED

Wide attribution tables still require horizontal scrolling on small screens, and long Vietnamese sector or industry labels can increase row height. Native SVG gives direct control over chart behavior, but the current implementation would benefit from broader screen-reader testing and user testing with investors outside the development team. The design system establishes consistency, but it does not prove that every statistical explanation is immediately understood.

The principal lesson is that visual design in a financial product protects meaning. Units, assumptions, source timing, and uncertainty are part of the interface rather than material to be added after calculation, and a visually prominent number can mislead when its period or basis is unclear. The strongest design decisions therefore tied hierarchy to evidence: result before interpretation, current allocation before alternatives, and methods beside the outputs that depend on them.

VI CONCLUSION

My contribution transformed Finfolio 2.0 from a collection of analytical functions into a coherent, multi-page user experience. I designed the information architecture, the Forest/Paper system, Vietnamese-compatible typography, actual-value chart behavior, attribution presentation, responsive layouts, and accessibility baseline.

The work maintained a clear accountability boundary. I was responsible for how verified outputs are ordered, labelled, and interacted with, while quantitative ownership remained with the financial and development roles. This separation allowed the final application to communicate complex portfolio evidence in a form that a retail investor can examine without losing sight of the underlying assumptions.

INDIVIDUAL DECLARATION

DECLARATION

I confirm that this report reflects the work that I am prepared to explain, calculate, demonstrate, or reproduce in Finfolio 2.0. Shared project artifacts are identified as collaborative where applicable, and my individual accountability is defined by the evidence presented in this report.

FULL NAME	Nguyễn Ngọc Anh
STUDENT ID	2413380008
SIGNATURE	
DATE	